

TCJastrowFitting

December 2025

To-Do List

- Format Alignment.
 - ✓ Set up the To-Do list.
 - Set up the format of the raw data.
- Depth-Adjusted ee term investigation.
 - ✓ Alpha interpolation between varmin and emin freely-optimized jastrow (ano-vdz/cc-pvdz/cc-pCVDZ/aug-cc-pvqz)
 - ✓ Freely-optimized varmin AE results with ee cutoff is set to be 6. (aug-ano-v{d/t}z)
 - ✓ Standard freely-optimized emin results of AE (ano-vdz/cc-pCVDZ/aug-ano-v{d/t}z/aug-cc-pvqz)
 - ✓ Standard freely-optimized varmin results of AE (ano-vdz/cc-pCVDZ/aug-ano-v{d/t}z/aug-cc-pvqz).
 - ✓ Ee-depth-fixed AE calculation (ano-vdz, aug-ano-vd/tz; Depth: 0.0, -0.1, -0.2, -0.3, -0.4, -0.5).
 - ✓ AE calculation with the ee depth fixed at the fitting-optimal depth. (aug-ano-vdz: Depth: -0.33,-0.37)
 - ECP: Fit quadratic functions for total energies and do the same panel plots like for AE.
 - ECP and AE: Plot difference of total energy of fixed depth and total energy of free floating depth such that we can compare the total energies of AE and ECP in a fair manner.
 - Plot the free floating e-e depths of AE Jastrows like Kristoffer did for ECPs.
 - Plot out the max deviation with MAE, RMSE.
- Depth-Adjusted en/een term investigation.
 - Scale the depth/height of en/een term in Jastrow.

- Spin-dependent Jastrow.
 - ✓ Standard freely-optimized varmin results of AE (Basis: aug-ano-v{d/t}z).
 - ✓ Opposite-spin ee-depth-fixed AE calculation (Basis: aug-ano-vd/tz; Depth: 0.0, -0.1, -0.2, -0.3, -0.4, -0.5).
 - Ee-cusp spin-dependent Jastrow.

Results

0.1 Free floating depths with varmin optimization (spin-independent)

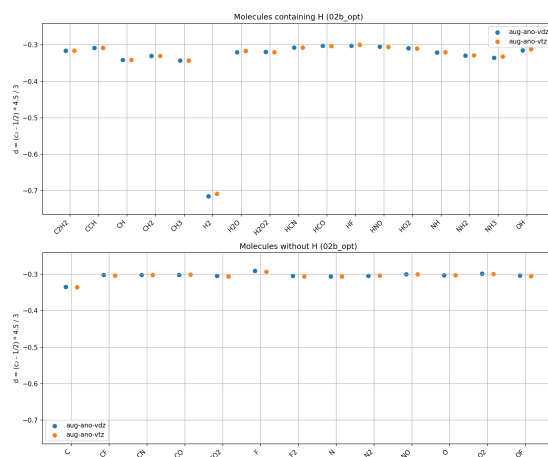


Figure 1: Free floating depths for AE calculations

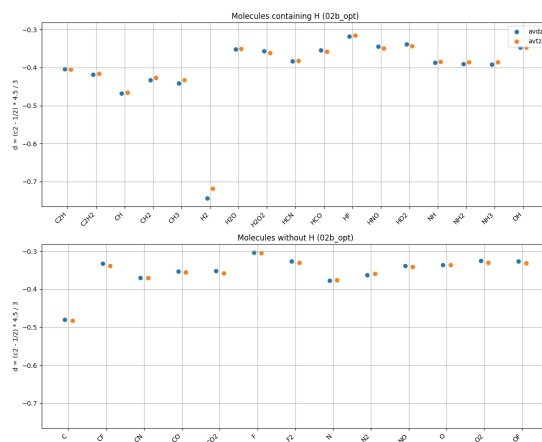


Figure 2: Free floating depths for ECP calculations

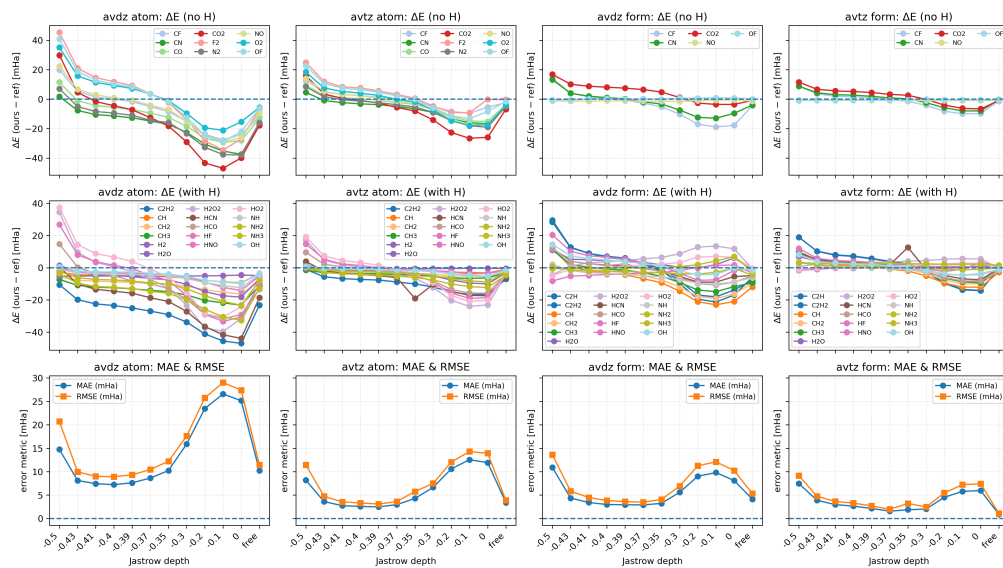


Figure 3: eCEPP

0.2 CCSD(T) energies against heat

0.2.1 eCEPP

0.2.2 ccECP

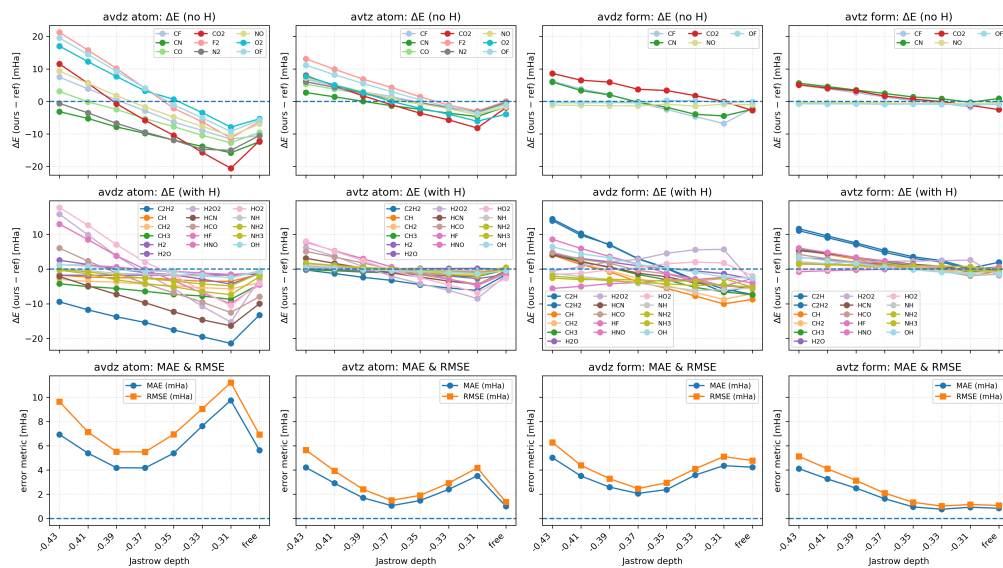


Figure 4: cECP

1 Introduction